

Matrix Notation for Derivatives

The Gradient Vector

The gradient vector of a real-valued function $f : \mathbb{R}^N \rightarrow \mathbb{R}$, i.e., $\nabla f(x) \in \mathbb{R}^N$, is used for optimization. Therefore, it must be a vector pointing in the direction you need to go so you can move up. It is a column vector that points in the direction of where the function increases the fastest, and its length is the rate at which the function increases.

The Derivative Matrix

The derivative matrix of a real-valued differentiable function $f : \mathbb{R}^N \rightarrow \mathbb{R}$, i.e., $Df(x)$, is a $1 \times N$ matrix; in fact, $\nabla f(x) = [Df(x)]^T$. $Df(x)$ is used for linearization, and it can be thought of as a “function.” Since this matrix doesn’t point in any direction, we use the directional derivative to see the rate of change of the function f when moving in a particular direction. Thus, we get a scalar for this. Important to note, the gradient vector is the *argmax* of the directional derivative.

The Chain Rule

Suppose that $g : \mathbb{R}^S \rightarrow \mathbb{R}^N$ and $f : \mathbb{R}^N \rightarrow \mathbb{R}^M$ are differentiable functions. The *composite* function $f(g(\cdot))$ is also differentiable. Consider any point $x \in \mathbb{R}^S$. The chain rule allows us to evaluate the $M \times S$ derivative matrix of the composite function with respect to x , $D_x f(g(x))$ by matrix multiplication of the $N \times S$ derivative matrix of $g(\cdot)$, $Dg(x)$, and the $M \times N$ derivative matrix of $f(\cdot)$ evaluated at $g(x)$, that is, $Df(y)$, where $y = g(x)$. Specifically,

$$D_x f(g(x)) = Df(g(x))Dg(x). \quad (\text{M.A.1})$$

The Product Rule

Here we simply provide a few illustrations.

(i) Suppose that $f : \mathbb{R}^N \rightarrow \mathbb{R}$ has the form $f(x) = g(x)h(x)$, where both $g(\cdot)$ and $h(\cdot)$ are real-valued functions of the N variables $x = (x_1, \dots, x_N)$ (Note: $g : \mathbb{R}^N \rightarrow \mathbb{R}$ and $h : \mathbb{R}^N \rightarrow \mathbb{R}$). Then the product rule of calculus tells us that

$$Df(x) = g(x)Dh(x) + h(x)Dg(x). \quad (\text{M.A.2})$$

which, transposing, can also be written as

$$\nabla f(x) = g(x)\nabla h(x) + h(x)\nabla g(x).$$

(ii) Suppose that $f : \mathbb{R}^N \rightarrow \mathbb{R}$ has the form $f(x) = g(x) \cdot h(x)$ where both $g(\cdot)$ and $h(\cdot)$ are vector-valued functions which map the N variables $x = (x_1, \dots, x_N)$ into $\mathbb{R}^M$. Then

$$Df(x) = g(x) \cdot Dh(x) + h(x) \cdot Dg(x). \quad (\text{M.A.3})$$

Note that $h(x) \cdot Dg(x) = [h(x)]^T Dg(x)$ is a $1 \times N$ matrix, as is the other term in the right-hand side. Thus, the vector-valued case (M.A.3) implies the scalar-valued formula (M.A.2).

(iii) Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}^M$ has the form $f(x) = \alpha(x)g(x)$, where $\alpha(\cdot)$ is a real-valued function of one variable (i.e., $\alpha : \mathbb{R} \rightarrow \mathbb{R}$) and $g : \mathbb{R} \rightarrow \mathbb{R}^M$. Then

$$Df(x) = \alpha(x)Dg(x) + \alpha'(x)g(x). \quad (\text{M.A.4})$$

(iv) Suppose that $f : \mathbb{R}^N \rightarrow \mathbb{R}^M$ has the form $f(x) = h(x)g(x)$ where $h : \mathbb{R}^N \rightarrow \mathbb{R}$ and $g : \mathbb{R}^N \rightarrow \mathbb{R}^M$. Then

$$Df(x) = h(x)Dg(x) + g(x)Dh(x). \quad (\text{M.A.5})$$

Note that $g(x)$ is an M -element vector (i.e., an $M \times 1$ matrix) and $Dh(x)$ is a $1 \times N$ matrix. Hence, $g(x)Dh(x)$ is an $M \times N$ matrix (of rank 1). Observe also that (M.A.4) follows as a special case of (M.A.5)